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cross-bridge cycles cause the sliding of the filaments only but there is no change in the lengths of either the thick or thin filaments. The Z- discs attached to the actin filaments are also pulled inwards from both the sides, causing the shortening of the sarcomere (i.e. contraction). This process continues as long as the muscle receives the stimuli and a steady flow of calcium ions. When motor impulse stops, the calcium ions are pumped back into the sarcoplasmic reticulum, results in the masking of the active sites of the actin filaments.The myosin head fails to bind with the active sites of actin and these changes cause the return of Z- discs back to their original position, i.e. relaxation.

Which myofilament has the binding sites for calcium? Name the specific molecule that binds with calcium.

Skeletal Muscle Glycogen Analysis (SMGA) –Used to measure an Athlete's sporting performance by taking muscle biopsies. It is a standard method to measure muscle glycogen. Muscle glycogen provides the main source of energy during anaerobic exercise. Furthermore, total glycogen stores within the body also contribute significantly to energy metabolism in endurance-type events lasting longer in duration. A single glycogen molecule may contain 5000 glucose units compared to that of 5000 individual glucose molecules.

Schematic Presentation of Muscle Contraction

Muscle contraction is initiated by the signal from CNS

↓

Release of acetylcholine at the neuromuscular junction

↓

Causes action potential in muscle fibre

↓

Triggers the release of calcium ions from sarcoplasmic reticulum

↓

Calcium ions combine with troponin and tropomyosin uncovers the binding sites on actin and initiates contraction

↓

Myosin binding sites on actin exposed. Myosin head binds to actin

↓

Myosin head executes power stroke

↓

Actin filament slides towards the centre of sarcomere (contraction)

↓

Signal from CNS stops; calcium ions are pumped back into the sarcoplasmic reticulum

↓

Tropomyosin masks the binding sites. Filaments pulled back to the original position (relaxation)

All muscles produce movement, but only skeletal muscle is responsible for locomotion. What is meant by this statement?